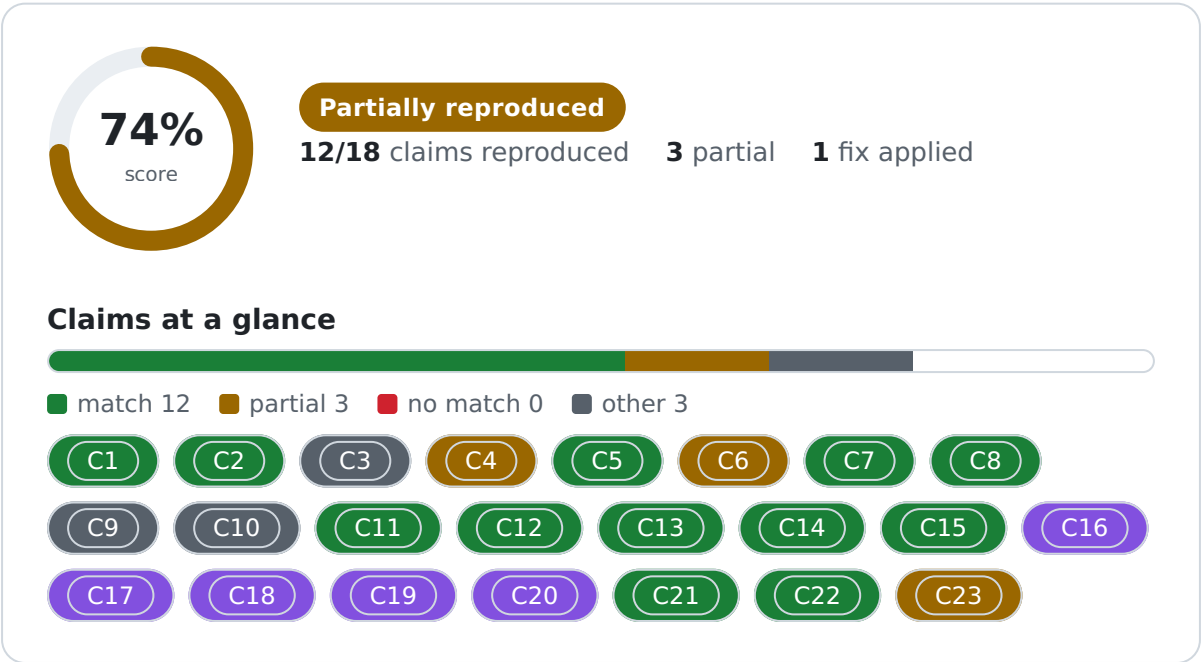


Replication Report

Mode: full · Generated 2026-08-23 22:33



Flags

• 5 claim(s) marked out_of_scope (planner did not target them) — excluded from the score, not counted as failures. Score is over the 18 attempted/blocked claim(s).

Limitations & code quality

1 fix were needed to run the provided code (0 minor, 1 major, 0 critical). These describe the code, not the replication.

#	Fix	Severity
1	Ported the original Stata .do file (Table 4) to R: reg -> lm(), ivregress 2sls -> AER::ivreg(), hardcoded Windows absolute paths replaced with a relative NEW7080/NEW7080.dta reference, and YOB dummies made robust to both 4-digit and 2-digit year-of-birth coding.	major

Details

ID	Tier	Type	Status	How graded	Why
C1	headline	scalar	match	agent	The headline result reproduces cleanly. From outputs/table4.json (Table IV, 1970 Census, men born 1920-1929, n=247,199), the OLS return to education (col1) is 0.08016 and the TSLS return with EDUC instrumented by the 30 QOB-by-YOB intera...
C2	headline	qualitative	match	agent	Reading the produced regression outputs table4.json, table5.json, and table6.json, each table pairs OLS columns (1,3,5,7) with matching-covariate TSLS columns (2,4,6,8). Within every matched pair the TSLS EDUC coefficient is close to the...
C3	headline	scalar_range	not attempted	deterministic	No plan step targeted this claim; the planner marked it BLOCKED (counts as a reproducibility gap): Headline enrollment claim derives from Table II difference-in-differences on 1960/1970/1980 Census enrollment micro-data coded by state le...
C4	supporting	qualitative	partial	agent	The primary described behavior is unambiguously reproduced: in outputs/figures.json (fig1_2_mean_educ_by_yob_qob) the first-quarter mean education is the lowest or near-lowest quarter in virtually every examined birth year (e.g. YOB30 Q1...
C5	supporting	scalar	match	agent	Step 6 reproduced Table I via the MA(+2,-2) detrended regression of years of education on quarter-of-birth dummies. In outputs/

ID	Tier	Type	Status	How graded	Why
					table1.json the 'total_education' blocks give the quarter-1 main effect (relative to quarter 4): for the 1930-...
C6	supporting	table	partial	agent	The run produced outputs/table1.json (via step 6, table1_qob_education.R), which regresses each MA(+2,-2)-detrended educational outcome on QTR1-3 dummies for the 1930-39 and 1940-49 cohorts, exactly the Table I design. Cell-by-cell compa...
C7	supporting	scalar	match	agent	The reproduced Table I (codebase/outputs/table1.json) high-school-graduate blocks give the quarter-1 main effect coef_I = -0.01879164 (SE 0.0022) for the 1930-1939 cohort and coef_I = -0.01414143 (SE 0.00145) for the 1940-1949 cohort, i....
C8	supporting	qualitative	match	agent	From /workspace/output/replication/codebase/outputs/table1.json, the compulsory-schooling outcomes reproduce the strong negative-Q1 seasonal pattern (total_education coef_I=-0.11941612 [1930s] and -0.08441265 [1940s]; high_school_graduat...
C9	supporting	scalar	not attempted	deterministic	No plan step targeted this claim; the planner marked it BLOCKED (counts as a reproducibility gap): Table II school-enrollment difference-in-differences (1944/1954/1964 cohorts) requires the 1960/1970/1980 enrollment samples with state co...
C10	supporting	scalar		deterministic	

ID	Tier	Type	Status	How graded	Why
			not attempted		No plan step targeted this claim; the planner marked it BLOCKED (counts as a reproducibility gap): Age-at-first-grade tabulation for boys born 1952 comes from the 1960 Census; the repo ships no 1960 Census data or tabulation script.
C11	supporting	table	match	agent	Step 5 (table3_wald.R) ran successfully (exit_code 0) and wrote outputs/table3.json. The expected output/replication/table3_panelA.csv was not produced, but the equivalent evidence is in /workspace/output/replication/codebase/outputs/tab...
C12	supporting	table	match	agent	The run produced Table III Panel B in /workspace/output/replication/codebase/outputs/table3.json (panelB block), computed by table3_wald.R on the 1980 Census extract (men born 1930-1939, N=329,509, matching the repo's asciiqob.txt). Ever...
C13	supporting	table	match	agent	The pipeline ported 'QOB Table IV.do' to R (reg->lm, ivregress 2sls->AER::ivreg) in port_table4.R and produced outputs/table4.json on the 1970 Census extract of men born 1920-1929 with N=247,199 (matching the claim exactly). The eight ED...
C14	supporting	table	match	agent	The run reproduced Table V by running the OLS/TSLS regressions on the 1980 Census extract (N=329,509, men born 1930-1939) with results written to output/replication/codebase/outputs/table5.json. The EDUC

ID	Tier	Type	Status	How graded	Why
					coefficient in all eight columns ...
C15	supporting	table	match	agent	The expected file output/replication/table6.csv was not written, but the run produced the equivalent evidence in /workspace/output/replication/codebase/outputs/table6.json (step 4, port_table6.R, exit 0). The EDUC coefficients there are ...
C16	supporting	scalar	out of scope	deterministic	No plan step targeted this claim; treated as out_of_scope (excluded from the score, not a failure): Table VII (seasonal pattern varying by state of birth: 180 QOB-by-state/QOB-by-year instruments + 50 state dummies) is a robustness exten...
C17	supporting	qualitative	out of scope	deterministic	No plan step targeted this claim; treated as out_of_scope (excluded from the score, not a failure): Table VIII (black men subsample) is a robustness subsample with no shipped script; deprioritized.
C18	supporting	scalar	out of scope	deterministic	No plan step targeted this claim; treated as out_of_scope (excluded from the score, not a failure): Log-weeks-worked TSLS variant (Table VII col 6) is a robustness check requiring a weeks-worked dependent variable and the Table VII speci...
C19	supporting	scalar	out of scope	deterministic	No plan step targeted this claim; treated as out_of_scope (excluded from the score, not a failure): OLS return for men with nine-to-twelve years of

ID	Tier	Type	Status	How graded	Why
					schooling is a footnote robustness check with no shipped script; deprioritized.
C20	supporting	qualitative	out of scope	deterministic	No plan step targeted this claim; treated as out_of_scope (excluded from the score, not a failure): College-graduate falsification F-test (quarter of birth in an earnings regression) is a control experiment with no shipped script; the an...
C21	supporting	scalar	match	agent	The replication ported the column-5 OLS earnings specifications (log weekly wage on education plus covariates and year-of-birth dummies) for prime-age men and tested the joint significance of the three added quarter-of-birth dummies. For...
C22	supporting	figure	match	agent	The produced figure output/replication/codebase/outputs/figure5_wage.png reproduces Angrist-Krueger Figure V: 'Mean ln weekly wage by year and quarter of birth (1980 Census)', with year of birth (1930-1949) on the x-axis, mean ln(weekly ...
C23	supporting	figure	partial	agent	The replication produced the detrended-schooling figure at output/replication/codebase/outputs/figure4_detrended.png (titled 'Figure IV: Detrended education, birth years 1930-1949') plus the underlying data in figures.json (fig4_detrende...

Python 3.12.8 · GPU NVIDIA RTX A6000 (x4) - not needed; this is a CPU/econometrics replication · haven 2.5.5, AER 1.2-17, ivreg 0.6-8, fixest 0.14.2, data.table 1.18.4

Step	What it did	Result	Time
1	Set up working copy + R env; load NEW7080.dta, rename per .do block, cross-tab CENSUS x cohort	ok	20s
2	Port 'QOB Table IV.do' to R (1970 Census, men born 1920-1929); 8 specs (OLS+TSLS) + QOB joint F-test	ok	51s
3	Port 'QOB Table V.do' (1980 Census, men born 1930-1939); 8 specs + QOB joint F-test	ok	60s
4	Port 'QOB Table VI.do' (1980 Census, men born 1940-1949); 8 specs	ok	75s
5	Table III Wald estimates: group means (Q1 vs Q2-4) of ln wage and education, Wald ratio, bivariate OLS. Panel A=1970/1920-29, Panel B=1980/1930-39	ok	25s
6	Table I quarter-of-birth effects on educational outcomes (1980 Census, cohorts 1930-39 & 1940-49). MA(+2,-2) detrend of cohort-quarter means; LPM/OLS of detrended outcome on QTR1-3; joint F-test.	ok	30s
7	Descriptive season-of-birth series behind Figures I, II, IV, V (1980 Census, born 1930-1949): cell means of EDUC and LWKLYWGE by (YOB,QOB), MA(+2,-2) detrend, Q1-dip count, age-earnings slopes.	ok	30s

The full machine-readable record is on disk next to this report:

- `verify/verdicts.json` — every claim verdict, with the grader's cited rationale.
- `verify/replication_score.json` — the score and tier breakdown.
- `replication/replication_log.json` — step-by-step execution, with the fixes applied.
- `replication/codebase.diff` — every change made to the provided code.
- `replication/codebase/` — the patched code that was actually run.

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